

Deep Multi-stage Model for Automated Landmarking of Craniomaxillofacial CT Scans

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Abstract—In this paper we define a deep multi-stage architecture for automated landmarking of craniomaxillofacial (CMF) CT images. Our model is composed of three subnetworks that first localize, on reduced-resolution images, areas where landmarks may be found and then refine the search, at full-resolution scale, through a hierarchical structure aiming at increasing the granularity of the investigated region. The multi-stage pipeline is designed to deal with full resolution data and does not require any additional pre-processing step to reduce search space, as opposed to existing methods that can be only adopted for searching landmarks located in well-defined anatomical structures (e.g., mandibles). The automated landmarking system is tested on identifying landmarks located in several CMF regions, achieving an average error of 0.8 mm, significantly lower than expert readings. The proposed model also outperforms baselines and is on par with existing models that employ additional upstream segmentation, on state-of-the-art benchmarks.

I. INTRODUCTION

Craniomaxillofacial (CMF) malformations due to abnormal development comprise over one third of all congenital birth defects [1], making high-efficacy CMF morphology assessment essential for identifying the most appropriate clinical treatment. To this aim, computer tomography (CT) and cone-beam computer tomography (CBCT) are the most common imaging modalities for diagnosis and treatment of CMF disorders. Standard clinical practices include visual inspection, followed by manual annotation, of CT images for precise anatomical landmarking. Landmarking is indeed one of the upstream steps for any subsequent clinical evaluation — e.g., surgical planning, deformity prediction and quantification — and, thus, localization accuracy is critical. However, accurate identification, beside being very time-consuming for physicians, is rather challenging as it is prone to various errors and subject to intra- and inter-operator variability. Hence, there is a strong unmet need for developing highly accurate and efficient landmarking methods to support clinical analysis.

Automated landmarking has consequently drawn significant attention in the years, but it is still an unsolved problem because of the high anatomical shape variability in CMF

images, although some recent approaches for accurate localization in CMF scans have been proposed [2], [3], [4]. One of the main limitations of the existing models is the need to have an upstream segmentation model to control landmark identification, which hinders their application to points (as those used in this work) lying on either unstructured or multiple structures/bones, for which it is not trivial to perform segmentation because some points are on mandibles, others are on airways, others on the vertebral column, etc. Furthermore, another major issue with landmarking is the need to work at full resolution scale, as tiny pixel errors may result in higher errors in volume space on resized data. However, training deep learning models — especially based on 3D convolutions — at full scale is not easy and requires either machines with dedicated and powerful hardware, or models that process data by patches, thus increasing the effective time for processing a single input.

Our approach addresses both problems through a three-level coarse-to-fine localization process that employs shared models for approximate landmark localization at resized scale, followed by landmark-specific models, operating a full scale, for more precise refinement.

To demonstrate the effectiveness of the proposed landmarking method, we test on two different sets of landmarks, namely, airways (i.e., nasal cavity and pharynx) to investigate OSAS and mandibles [4] for CMF analysis, showing, in both cases, promising performance and generalization capabilities.

II. RELATED WORK

Before the advent of deep learning, CMF bone segmentation and landmarking were mainly addressed through atlas-guided methods [5], [6], [7], [8], whose performance, however, depends significantly on the template registration phase, thus failing in case of CMF bone deformities.

It is known that deep learning has had a notable impact on the medical images analysis domain, in general, and in CMF CT scan analysis in particular. However, while deep learning-based automated segmentation of CMF bones and

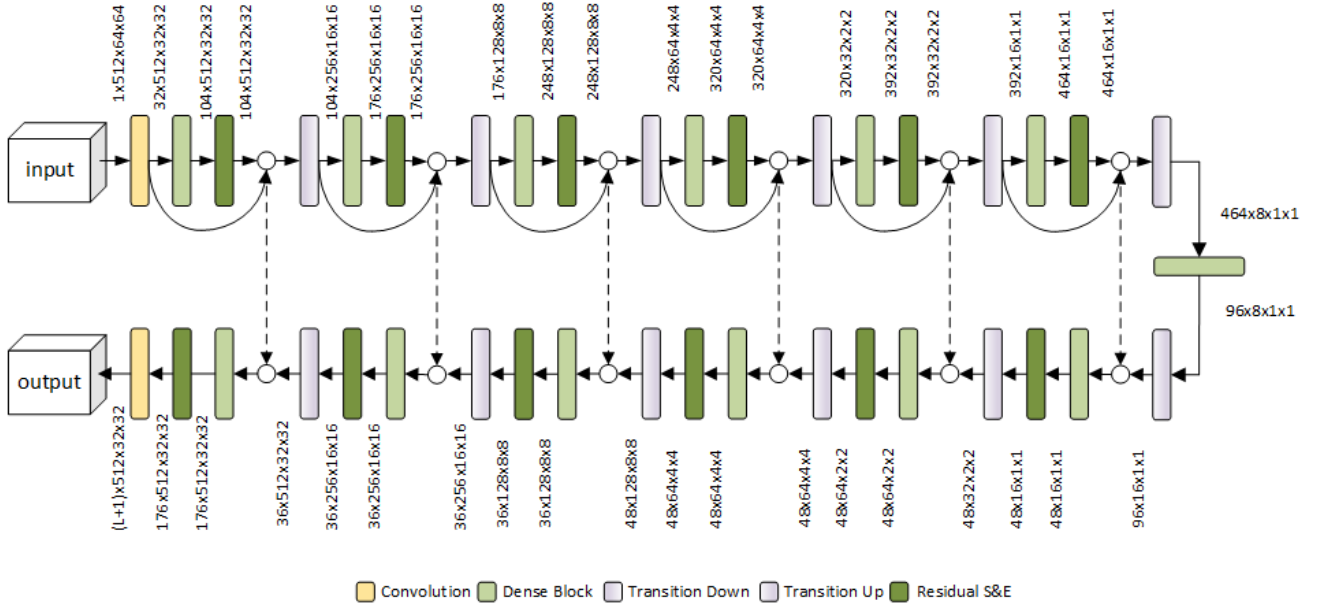


Fig. 1. Landmark plane localization model.

structures has been largely investigated with very promising results, mainly relying on the U-Net architecture [9] and its variants [10], [11], [12], [13], automated landmarking is still far from reaching the same level of accuracy and flexibility. Recent models have attempted to identify landmarks through joint CMF segmentation and landmark identification [14], [4] by leveraging the high segmentation accuracy to control better the landmark search space. For example, [4] describes an approach for joint mandibular segmentation and landmarking; segmentation masks, extracted through [15], are used for learning a geodesic space for subsequent landmark detection. However, all these methods require an upstream segmentation step to reduce the search space for landmarks. While this might be an effective solution for landmarks lying on well defined surfaces, such as mandibles, it might fail, instead, for landmarks located on less-structured surfaces, e.g. airways. In this paper we face this problem through a multi-stage pipeline that does not employ any prior segmentation and searches for landmarks at full CT resolution. This makes our system able to work on unstructured surfaces, such as sino-nasal or pharyngeal airways, where existing solutions are still based on traditional computer vision techniques [16], [17].

III. LANDMARK LOCALIZATION

The proposed model operates a coarse-to-fine localization process by employing a hierarchy of three models, which sequentially process a CT 3D scan to identify first the slice (along the x axis¹) where a landmark should be found, then the approximate region within that slice (a window in the yz plane), and finally the precise location inside the region (the (y, z) coordinates). The first two models process the

¹Note that the reference xyz reference system is arbitrary: we specify axis names for readability and consistency.

whole CT scan and are shared in the detection process of all target landmarks; the third model is landmark-specific, but less computationally demanding as it only processes a small 2D region — at full resolution — containing the target location. The possibility to search landmarks at full resolution in a limited area allows our model to significantly reduce the misplacement error, as shown later.

Landmark plane localization. The model designed to identify the location of the x coordinate for each target landmark is a 3D variant of the Tiramisu architecture [15], obtained by replacing 2D convolutions with 3D convolutions. The input to the model is a $512 \times 512 \times 512$ CT scan downsampled along the y and z dimensions, thus producing a $512 \times 64 \times 64$ volume where maximum detail is kept for the axis along which we need to identify the correct plane for each landmark. Such downsampled input is then processed by a sequence of dense blocks, residual squeeze-and-excitation layers and transition-down layers (the details are given in Fig. 1), in order to obtain a $(L + 1) \times 512 \times 32 \times 32$ output, with L being the number of landmarks to identify and an additional channel accounting for background likelihood. We treat the network as a voxel-wise classification model, estimating for each pixel the likelihood of being either background or one of the L landmarks. For each landmark $l \in \{1, \dots, L\}$, we then select its x_l coordinate given the log-likelihood tensor P as follows:

$$x_l, y_l, z_l = \arg \max_{x, y, z} P(l, x, y, z). \quad (1)$$

Simply put, we compute the x coordinate of the largest-scoring voxel of the l -th log-likelihood volume. At this stage, y_l and z_l are ignored.

Landmark region localization. This model (shown in Fig. 2) identifies a small area on the yz plane around each candidate landmark location. Employing an intermediate model for

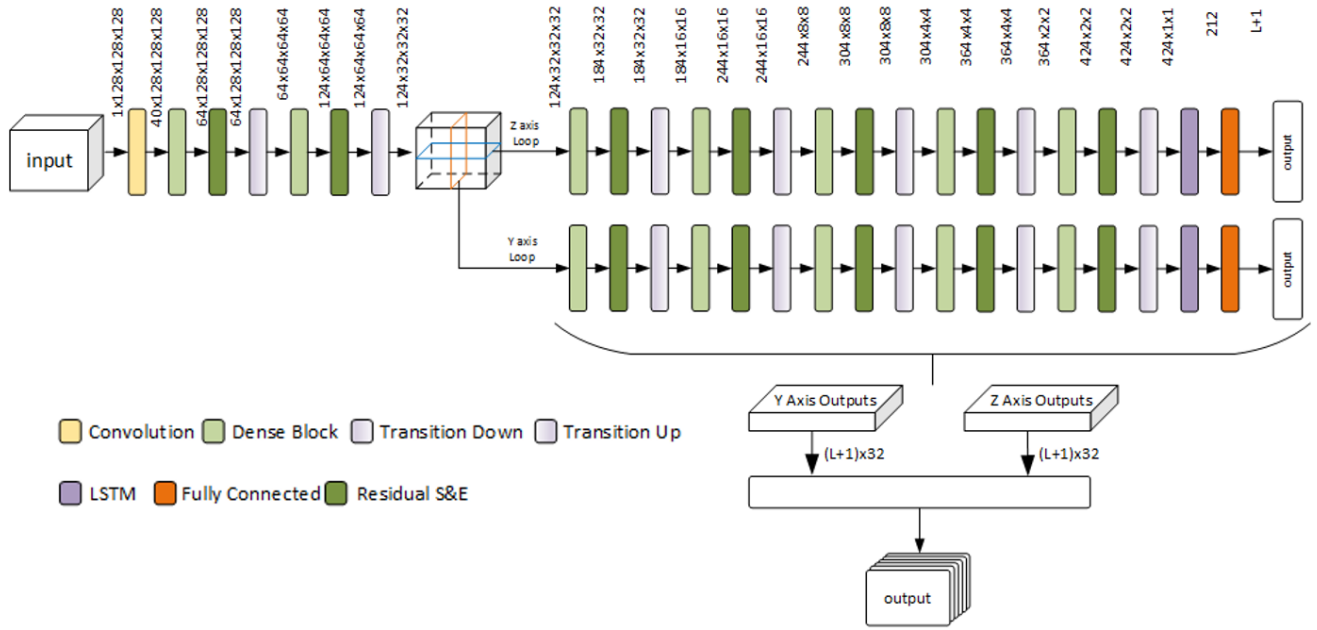


Fig. 2. Landmark region localization model.

region localization, rather than directly computing landmark positions, enables to process that predicted region at its full resolution in the following stage, thus increasing the accuracy of the final estimated landmark locations. Additionally, since a single pass of the model estimates the regions surrounding all landmarks, the computational complexity of such model is still constant with respect to the number of landmarks. The input to the model is a CT scan resized to $128 \times 128 \times 128$. A cascade of dense blocks, residual squeeze-and-excitation and transition-down layers further reduces the spatial size to $32 \times 32 \times 32$. At this point, the model forks into two branches, each processing one single slice at a time (one branch processes slices along the y axis, while the other processes slices along the z axis). The two branches are structurally identical, but process 2D slices of the input feature volume along the y and z axis separately, in order to estimate the corresponding coordinates (i.e., the center of the predicted region) independently.

Within each branch, a cascade of 2D dense blocks, residual squeeze-and-excitation and transition-down layers process a 32×32 plane of the feature volume by reducing it to a 424-dimensional vector and obtaining a sequence of 32 such vectors (one per slice). A downstream 1D bidirectional LSTM (for each branch), with state size of 212, processes all slices. The resulting output state sequence encodes directional spatio-temporal information, which is then mapped to an $(L+1)$ -vector of class scores for background and landmarks. The two branches thus provide a $32 \times (L+1)$ matrix each, representing class distributions at the corresponding binned coordinates. Then, for each class, we view the corresponding spatial distributions as two 32×1 and 1×32 matrices, whose matrix product provides a 32×32 matrix where each location encodes the score of finding a given landmark (or

background). Concatenating all 32×32 class matrices into a single volume, and employing Eq. 1, we can estimate the approximate (due to pooling) y and z coordinates of each landmark and extract a patch of size 160×160 around it.

Landmark point localization. The analysis carried out by the two previous models provides, for each target landmark, the corresponding x coordinate and a 160×160 search region on the $y-z$ plane. Up to this point, a single network for each of the previous two steps is employed to estimate approximate coordinates for all target landmarks. In order to refine these predictions and compute pixel-accurate coordinates of each landmark, we finally train separate models that process the search regions returned in the previous stage. The architecture of the model (shown in Fig. 3) is based on a sequence of proposed blocks (i.e., dense block followed by residual squeeze-and-excitation and transition-down layers) and estimates a probability map of each pixel being the landmark location. The maximum of this map is taken as the (y, z) coordinate, that is combined to the initially-estimated x coordinate to provide the final landmark location.

IV. EXPERIMENTS AND RESULTS

Datasets and Metrics. In our experiments for landmark detection, we employ the following two datasets:

— **AirwaysSet** containing of 19 CT scans of 19 patients. Scans were performed using a standardized scanning protocol using an iCAT Next Generation CBCT unit. Each CT scan is used at its full resolution, i.e., $512 \times 512 \times 512$. The dataset contains the locations (as voxel coordinates) of five landmarks of clinical interests for airways analysis: Anterior nasal spine (ANS), Posterior nasal spine (PNS), Superior-posterior

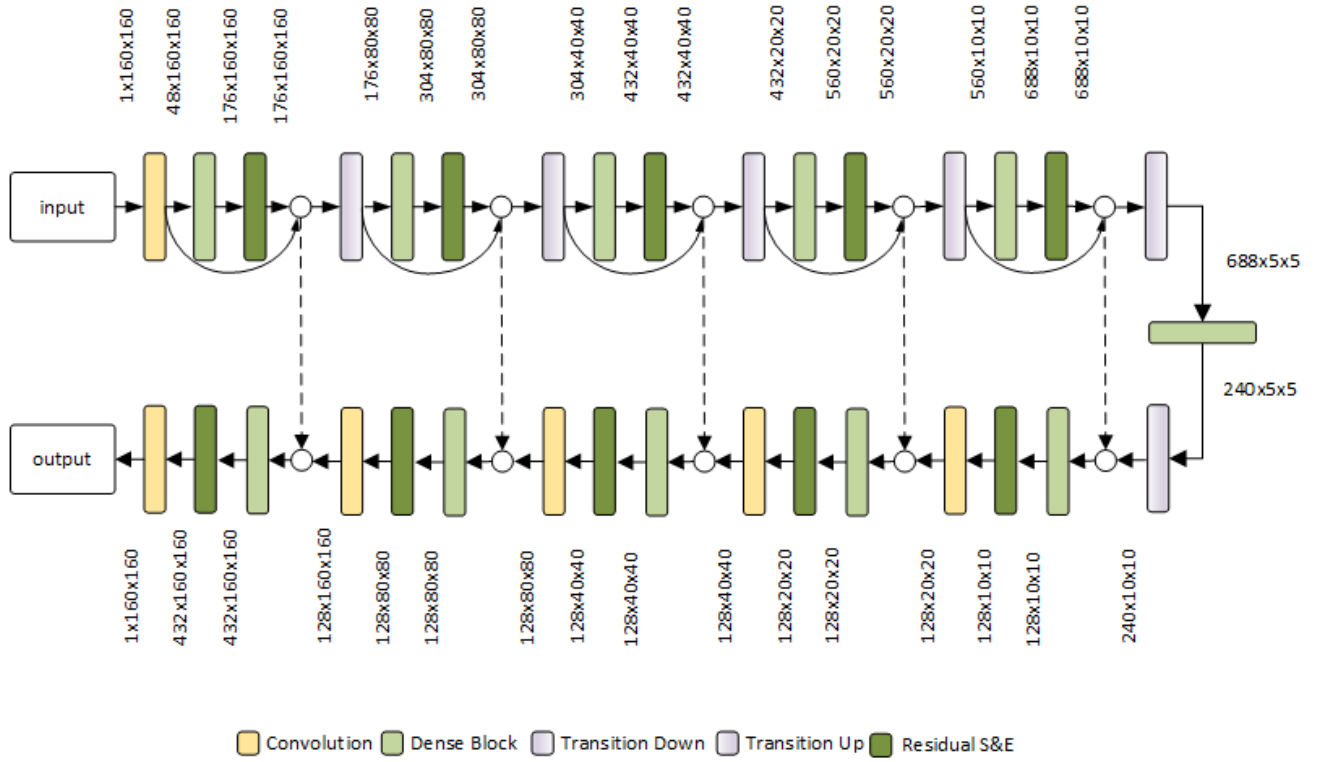


Fig. 3. Landmark point localization model.

extremity of the odontoid process of C2 (*C2sp*), Anterior-inferior extremity of the body of C3 (*C3ai*), Nasion (*N*). The landmarks are generally employed to study obstructed sleep apnea syndrome (OSAS). Manual localization was done by two expert radiologists using Mimics for the annotation process.

— **Dataset in [4]** comprises anonymized CBCT scans of 50 patients. All images were obtained on a Hitachi CB MercuRay CBCT system. Each CBCT scan comes with the volume coordinates of 9 cephalometric landmarks: Menton (*Me*), Gnathion (*Gn*), Pogonion (*Pg*), B Point (*B*), Infradentale (*Id*), Condylar Left (*CdL*) and Right (*CdR*), Coronoid Left (*CorL*) Coronoid Right (*CorR*). These landmarks are adopted in clinical routine for planning CMF surgery or for deformity quantification.

Landmarking performance is reported as the detection error (in mm), averaged through 5-fold cross-validation.

For training, we apply a voxel-wise cross-entropy loss; in the plane localization and point localization models, we weigh background voxels by a factor of 10^{-5} , in order to compensate the higher relative frequencies with respect to landmark voxels. We employ RMSProp as optimizer, with learning rate of 0.0001, weight decay of 0.0001, and batch size of 2. All code, to be released, is written in PyTorch 0.3; experiments executed on an nVidia Titan X (Pascal).

To test our model on [4]’s dataset, we pre-train our models on the Airways Dataset and fine-tune them on [4], using again 5-fold cross-validation and the same training setup.

Landmark Localization results. We first evaluate the performance of our landmark localization model, including comparison with proper baselines, on the *AirwaysSet* dataset. Results are computed in terms of localization errors in mm and are shown in Tab. I. As baseline we used a model that operates with resized (to $128 \times 128 \times 128$) data. The baseline architecture is based on the standard Tiramisu, with 3D convolutions instead of 2D ones. The $128 \times 128 \times 128$ input is first fed to a $5 \times 5 \times 5$ convolution layer, which does not increase the number of channels, and is then processed by 7 dense blocks with 2, 2, 3, 4, 5, 5, 5 convolution layers respectively (with $3 \times 3 \times 3$ kernels). At the end of each dense block, max pooling is applied to the feature maps to reduce dimensions. The downsampling path is followed by a bottleneck dense block with 16 $3 \times 3 \times 3$ convolution layers. The structure of the upsampling path is symmetrical to the downsampling path and is followed by a $5 \times 5 \times 5$ convolution layer whose output is a $128 \times 128 \times 128$ tensor. The growth rate used in all of the dense blocks is 8. The choice of the baseline (and its input data) is made in order to allow training on our GPU and to show the effectiveness of our multi-stage approach in searching landmarks in a high resolution space. Results (given in Table I) show an average volume error of 0.85 mm, significantly lower than the baseline (which is 2.12 mm). The main difference is achieved on point C3ai (1.06 vs 5.80), as this is a challenging point requiring full resolution to be found precisely. Qualitative examples of landmark detection is shown in Fig. 4.

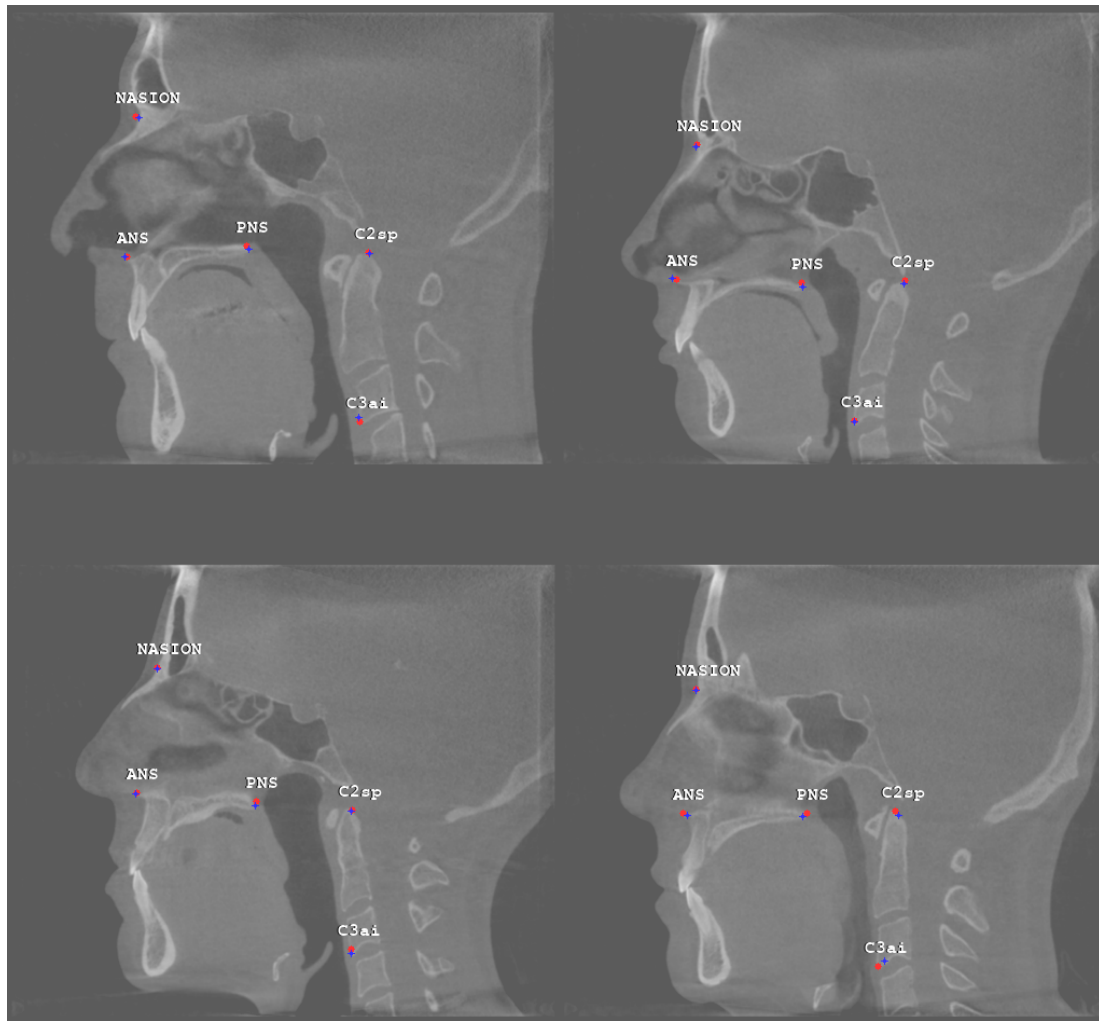


Fig. 4. Examples of landmark detection. Blue dots show the landmarks localized by the two radiologists (average between the two sets of measurements), while red dots are the landmarks automatically identified by the proposed model.

	Our model	Baseline
ANS	0.88 ± 0.64	1.49 ± 1.11
C2sp	0.74 ± 0.50	1.01 ± 1.01
C3ai	1.06 ± 0.83	5.80 ± 4.91
N	0.57 ± 0.53	1.23 ± 1.48
PNS	1.01 ± 0.56	1.09 ± 0.77
Average	0.85	2.12

TABLE I
LANDMARK LOCALIZATION ERROR (IN MM) ON *AirwaysSet*.

Note that on our *AirwaysSet* dataset the plane and region localization models can benefit from the co-planarity of the target landmarks, which lie on a single plane parallel to the sagittal plane. To exploit this property, the x axis is chosen to be perpendicular to the sagittal one, so that landmark-shared features can be computed by the first plane localization model. This property will not hold for the dataset in [4], which thus provides a more general test example.

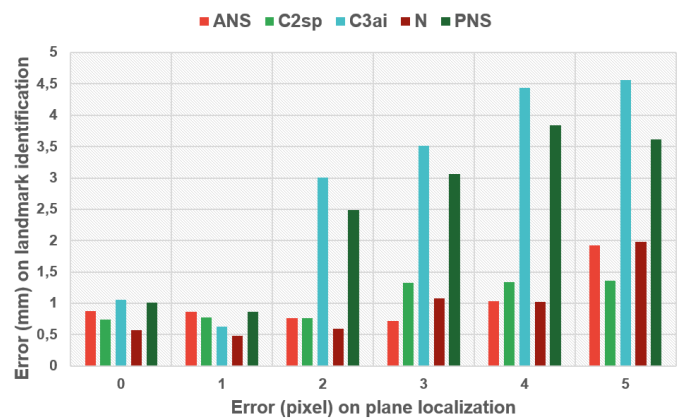


Fig. 5. Impact of localization error (in pixel) on plane localization model on the final landmark localization accuracy. 1 pixel error corresponds to 0.4 mm error.

Since our approach is based on a multi-stage architecture, we evaluate error propagation from upstream to downstream

	Our model	Torosdagli et al [4]
Me	0.84	0.33
Cor _L	0.71	0.27
Cor _R	0.52	0.03
Cd _L	1.08	1.01
Cd _R	0.47	0
Gn	0.91	0.41
Pg	0.93	1.36
B	0.73	0.68
Id	0.85	0.35
Average	0.78	0.55

TABLE II
COMPARISON WITH STATE OF THE ART IN TERMS OF LOCALIZATION
ERROR (IN MM). VALUES FOR [4] ARE TAKEN FROM THEIR PAPER.

models. More specifically, we investigate what is the effect of misjudgement in plane localization on the final landmark assessment. We do not carry out the same analysis on the region localization model since it does not estimate exact coordinates, but only has the effect to reduce the memory occupation in the last module. Fig. 5 reports the landmark localization accuracy as we induce a random error (from 0 to 5 pixels) on the plane localization model. Results show that: 1) our plane localization model is quite accurate, as a small offset of 2 pixels leads to significant lower performance (indeed, average plane localization error in our case is 1.42 mm); 2) C3ai and PNS are the landmarks most affected by the plane localization accuracy (which we also observe when using the baseline), highlighting the importance of the multi-stage approach.

We also compare the landmark localization error to overall expert reading variations. We observe that, while variation among experts is about 2 mm (which is clinically acceptable), our approach obtains an average error of 0.85 mm, with the highest error (for C3ai landmark) of 1.06 mm.

As a final evaluation, we also compare our approach to the recent landmarking model in [4] on their dataset. In this case, we first pre-train our model on the *Airways dataset*, then fine-tune it on the target dataset. Results are given in Tab. II and show that our approach yields an average landmarking error of 0.78 mm, which is almost on-par with [4], i.e., 0.55 mm. However, it should be noted that our approach does not require any prior segmentation. This is a clear advantage w.r.t. [4], as it cannot be applied to points, as in the *AirwaysSet*, for which it is hard to apply ad-hoc segmentation procedures. Indeed, the considered points are located in different structures or in joints between them: nasion (N) is located in the frontonasal suture, ANS and PNS are located in the nasal spine, C2sp and C3ai are located in joints between airways and, respectively, C2 and C3 vertebrae. Thus, the proposed approach yields better generalization capabilities and it can be applied to different types of landmarks.

V. CONCLUSION

In this paper we propose a learning-based approach for automated landmarking of CT images. While traditional methods

are forced to reduce the search space by controlling the search space through segmentation masks, our multi-stage model is able to work directly on the input data at its full resolution. This leads to an average landmarking error of less of 1 mm on two different CMF structures (mandibles and airways), which is significantly lower than the normal readings by experts and on-par with state of the art methods [4] that, however, need a segmentation module for each investigated CMF structure.

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